TECHNICAL DOCUMENTATION

APPLIED AI · RAG

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Riansares 4R — Patio, Home, Damp and Repair Cases

REFERENCE DOCUMENTATION FOR THE AI ASSISTANT

RIANSARES 4R

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6. Case A — Courtyard, Base, Paving and Drainage

6.1. Context

Case studied in a house in Uclés with:

- exterior courtyard;
- side corridor;
- interior courtyard;
- partial new bases;
- exterior tiles;
- welded wire mesh;
- different access conditions;
- inspection chambers, downpipe, gutter and drain grate;
- need for slopes and water evacuation.

The central lesson was that “laying a floor” does not describe the actual work. There is preparation, base, concrete, logistics, times, slopes, drainage, joints, and finishes.

6.2. Confirmed Measurements of the Case

Approximate final paving area:

Zone	Final Paving
Exterior Courtyard	29,03 m²
Corridor	8,94 m²
Interior Courtyard	22,43 m²
Total	**60,40 m²**

Approximate new base area:

Zone	New Base
Exterior Courtyard	16,07 m²
Corridor	8,94 m²
Interior Courtyard	18,17 m²
Total	**43,18 m²**

The 60,40 m² of tiles must not be mixed with the 43,18 m² of base.

6.3. Distinction Between Concrete Floor Base and Floor Finish

Concrete floor base: prepared concrete or mortar base to receive the paving, with preparation, thicknesses, reinforcement, slopes, and joints.

Floor tiling: placement of the final paving on an appropriate base.

A budget may have:

- fewer m² of base than paving;
- different concreting methods;
- the same type of tile on different supports;
- non-comparable costs per m² if mixed items are combined.

6.4. Concrete from Truck and Manual Execution

Exterior Courtyard with Access

Advantages of truck:

- close discharge;
- rapid execution;
- homogeneous mix;
- less handling;
- lower risk of concrete setting before spreading;
- fewer days of wheelbarrow and concrete mixer work.

Interior Courtyard or Narrow Corridor

Difficulties:

- indirect access;
- transportation with wheelbarrows;
- mixing on-site or phased pouring;
- heat and loss of useful time;
- welded wire mesh and spacers installed in phases;
- junctions and gradients;
- more labor;
- higher risk of poor execution if too much is attempted at once.

Reusable conclusion:

The paving may vary little between zones; the major difference is usually in how the base is executed.

6.5. Simplified Editorial Example

Example to explain to the client, not a universal rate:

- 60 m² of final flooring;
- 40 m² of new concrete floor base;
- 15 m² of exterior concrete floor base with good access;
- 25 m² of interior concrete floor base or manual.

Didactic result used:

- exterior concrete floor base: around 65 €/m² in the example;
- interior/manual concrete floor base: around 100 €/m²;
- exterior flooring: around 45 €/m²;
- interior flooring: around 50 €/m².

The comparison shows that cost does not depend solely on the material. It depends on the method, access, and working hours.

6.6. Corrected Budget Error

In the case, a localized waterproofing cost of 350 € was mistakenly reintroduced and should have been excluded from the adjusted version. The correction reduced the budget from around 7.450 € to around 7.090 €.

Editorial lesson:

- correct with transparency;
- explain which item entered by error;
- maintain the technical recommendation if it remains valid;
- do not dramatize;
- distinguish “recommended” from “included”.

6.7. Drainage and Slopes

Topics to develop:

- gradient towards inspection chambers or drainage channels;
- low points;
- intersections with facades and doors;

- external grates;
- downspouts and gutters;
- contraction joints;
- perimeter joints;
- compacted base;
- thickness and reinforcement;
- water trapped;
- dampness due to poor patio design.

7. Case B — Old House: Rehabilitate Without Excessive Demolition

7.1. Context

Old rural two-story house in the province of Cuenca. The following were observed:

- dampness in walls and ceilings;
- peeling finishes;
- old tiles and flooring;
- aged plumbing;
- poorly functional kitchen;
- narrow bathroom;
- windows with low insulation;
- chimney pending adaptation to a pellet stove;
- furniture and objects that hinder the work.

The case evolved from a broad renovation with layout changes toward a more selective proposal.

7.2. Principle: Negotiate with the Existing Structure

Moving partitions or creating openings in an old house may involve:

- structural risk;
- reinforcements;
- changes to plumbing and drainage;
- more demolition;
- more debris;
- more uncertainty;

- possible licenses or technical expertise;
- loss of time and budget in changes that add little value.

Recommended strategy:

- maintain the general layout;
- concentrate investment in the kitchen;
- improve the bathroom within its footprint;
- renovate priority installations;
- strip back and repair walls and ceilings;
- repair floors by zones;
- replace deficient windows;
- improve comfort and habitability.

A selective renovation is not a worse renovation. It can be more sensible and cost-effective.

7.3. Humidity in Old Walls

The following can coexist:

- Rising damp;
- Lateral water ingress from ground or rock;
- Condensation;
- Historic water ingress;
- Leaks;
- Salts.

Prudent interior process:

1. Remove loose finishes;
2. Chisel until sound substrate is reached;
3. Clean salts and dust;
4. Reconstruct losses in layers;
5. Use compatible and breathable mortars, preferably mineral or lime-based when applicable;
6. Finish with a compatible paint;
7. Explain that interior treatment does not eliminate an active water entry point.

Do not generally cover a damp or salt-affected wall with conventional plaster.

7.4. Hidden Damage

When chiseling, the following may appear:

- Sound bricks;
- Disintegrated bricks;
- Washed-out mortar joints;
- Weak old repairs;
- Cavities;
- Greater thickness than expected.

Clause of criteria:

Significant hidden damage is documented, explained, and budgeted before proceeding.

7.5. Small Bathroom

Criteria:

- Renovate within the existing space;
- Renew old plumbing;
- Waterproof the shower;
- Inspect the shower tray, drain and seals;
- Use a compact sink;
- Evaluate an outward-opening door or sliding door;
- Move the toilet only if the waste pipe and slope allow it;
- Do not assume that microcement is cost-effective.

Microcement is a thin coating over a stable, waterproof base. It does not replace a defective base.

7.6. Kitchen

The kitchen was considered the room that contributed the most value:

- Carefully chosen budget or lower-mid-range units;
- High cabinets up to the ceiling;
- Closed storage;
- Simple countertop;
- Adaptation of outlets and lighting;
- Preserve water and drain positions if feasible;

- Separate appliances unless agreed otherwise.

Commercial message:

We do not sell only furniture; we resolve a disorganized, hard-to-clean, and poorly functional kitchen.

7.7. Installations

Priority:

- Renew old pipes;
- Inspect drains;
- Preserve locations when feasible;
- Adapt electricity in renovated areas;
- Do not perform a full electrical renovation without a diagnosis.

Renovating finishes without replacing a very old pipe may require reopening the area.

7.8. Floors

Treat by zones:

- Lift where loose or cracked;
- Open where installations pass;
- Preserve stable support;
- Place tile over tile only if the base is firm, clean, flat, and a compatible system is used;
- Evaluate vinyl only after reviewing moisture and flatness.

"Tile over tile" is not automatically a mistake. The key is the substrate and preparation.

7.9. Rural Logistics

Include:

- Accommodation;
- Fuel;
- Travel;
- Local purchases;
- Tools;
- Material delivery;
- Access for a truck or container;
- Manual removal of debris.

Logistics can consume the margin of an apparently profitable project.

8. Case C — Communal Entrance with Damp and Old Substrates

8.1. Context

Entrance hall of a small owners' association in an old building in Tarancón, with:

- thick walls;
- arch of about 60 cm;
- finishes from various periods;
- old gypsum plasters and mortars;
- tiles;
- panels, meters, mailboxes, and wiring;
- peeling paint and finishes;
- prior treatment for moisture by a specialized company.

The original cause of the moisture was not confirmed.

8.2. Previous Treatment

A different mortar was observed at the lower part, apparently hard and stable. Verbally, it was indicated that tiles could be reinstalled leaving a small open band at the base.

The exact product is unknown.

Rule:

- conserve the treatment if it is firm;
- check hardness, adhesion, hollow sound, blistering, salts, and visible moisture;
- stop and review if the substrate fails;
- do not claim which system was used without documentation.

8.3. Geometry and Measurements

The main entrance is a clear rectangular opening. The stairs begin outside the rectangle, to the right.

Recorded data:

- approximate length of the portal: 6,00 m;
- front of the entrance: 3,00 m;
- general height: 2,50 m;
- maximum height at the rounded junction: 2,55 m;

- adopted equivalent work surface: about 22 m².

The equivalent work surface incorporates difficulty of:

- arch;
- returns;
- curves;
- meters;
- stairs;
- finishes;
- changes of substrate.

Do not value solely by planar square meters.

8.4. Recommended Mineral Solution

Sequence:

1. check existing substrate;
2. conserve what is firm;
3. chip hollow or loose parts;
4. remove isolated tiles only where they interfere;
5. clean dust and salts;
6. rebuild deep losses by layers;
7. place alkali-resistant mesh at substrate transitions;
8. regularize with rehabilitation mortar or breathable mortar;
9. reproduce compatible texture;
10. paint with a breathable system;
11. protect installations and remove debris.

Advantages:

- rehabilitates the substrate;
- maintains better breathability;
- does not reduce width;
- adapts to arches and curves;
- allows seeing and repairing the wall.

Limitations:

- more chipping;
- more debris;
- more drying;
- greater work;
- does not eliminate the external cause of moisture.

8.5. Wall Lining Alternative

Do not bond boards directly to a wall with risk of moisture.

Studied system:

- independent wall lining;
- independent metal framing;
- chamber;
- moisture-resistant board;
- separation from the floor;
- two grilles spaced apart, preferably one low and one high;
- mineral repair in arches, curves and junctions.

Advantages:

- less demolition on the main wall;
- regular finish;
- less debris;
- faster execution in straight panels.

Limitations:

- hides the wall;
- reduces space;
- does not solve the origin;
- requires chamber and ventilation;
- complicates meters and junctions;
- curved areas still require mortar.

8.6. Plaster, mesh and texture

- Conventional plaster may be the weak link in a damp and salt-affected substrate.

- A primer does not convert deteriorated plaster into a reliable substrate.
- Mesh is useful in transitions; it does not cure structural movements.
- A stippled texture (gotelé) may visually integrate and mask minor irregularities, but does not replace proper regularization.

8.7. Warranty and exclusions

In the case, a limited execution warranty was recommended, excluding:

- pre-existing moisture;
- leaks;
- drips;
- condensation;
- building movement;
- substrate defects;
- failures of third-party treatments;
- active water entry;
- lack of maintenance or clogged grilles.

The warranty must cover defects attributable to execution, not unresolved external pathologies.

9. Case D — Stained ceiling after a repaired leak

9.1. Context

Duplex residence with upper bathroom and lower ceiling with:

- old patches;
- circular stains;
- cracks around repairs;
- texture differences;
- existing bathroom exhaust fan but disconnected or disabled.

The first hypothesis was an active leak. Later it was confirmed that the leak had already been repaired.

9.2. Main lesson

Do not demolish before confirming the current condition.

Correct order:

1. ask if the leak is still active;

2. confirm what was repaired;
3. measure moisture;
4. check if the plaster or paint are hollow;
5. strip back and repair only what is necessary;
6. block the stains;
7. paint the entire surface;
8. expand scope only with authorization.

9.3. Ceiling repair

Process:

- protect floor and furniture;
- scrape loose paint;
- remove poorly adhered putty or plaster;
- open cracks when necessary;
- use mesh in relevant cracks;
- regularize;
- apply a substrate consolidant if the surface is powdery;
- apply stain-blocking primer;
- paint the entire ceiling or defined surface;
- apply the necessary coats.

Do not confuse:

- anti-humidity paint;
- stain blocker;
- finish paint.

Paint does not repair a leak. In this case the leak was already fixed and the goal was to restore the finish.

9.4. Why paint the entire ceiling

Painting only the circles often leaves:

- differences in whiteness;
- texture changes;
- different gloss;

- visible edges;
- a patchy appearance.

That is why it is advisable to paint the entire surface when feasible.

9.5. Existing exhaust fan

Before buying a new one:

- check wiring;
- measure voltage;
- inspect the switch;
- clean the unit;
- check if it rotates;
- inspect the duct and grille;
- replace only if faulty.

The base budget must exclude:

- new exhaust fan;
- new electrical line;
- service chases;
- duct repair;
- hidden malfunctions.

10. Case E — Old Toilet, Cistern and Minimum Viable Repair

10.1. Context

Old installation with:

- aged angle valve;
- rigid metal pipe;
- oxidation;
- awkward access;
- exposed cistern;
- components with possible incompatibility.

The first action replaced the valve and supply. The cistern remained pending.

10.2. Components That Should Not Be Confused

Angle Valve

Cuts off the supply from the wall.

Likely visual configuration of the case:

- inlet 1/2";
- outlet 3/8".

Must be noted as an inferred, unmeasured component.

Flexible Hose

Connects the valve to the cistern. Must:

- not be twisted;
- not be under tension;
- not form an overly tight curve;
- allow for maintenance.

Flush Mechanism

Opens the water outlet toward the bowl.

Push Button

Is the visible actuation, not the entire mechanism.

Fill Valve or Float

Allows water to enter and cuts off when the level is reached. It is not the flush valve.

Floor Fixation

Fixes the bowl to the floor. It does not connect the cistern and bowl.

Cistern-Bowl Screws and Seal

Connect both components and may be oxidized or hardened in an old sanitary fixture.

10.3. Risk of a Small Repair

An apparently simple task can become complicated due to:

- corrosion;
- seized components;
- dubious threads;

- fragile ceramic;
- hardened seals;
- improperly sized flexible hoses;
- lack of compatible replacement parts.

The cost should not be reduced to "ten minutes of work." It includes diagnosis, purchase, travel, risk, and warranty.

10.4. Reducing Scope

The client had planned to replace the toilet months later. Therefore, the scope was reduced to stopping the loss and restoring the filling, without renewing fixings or disassembling more than necessary.

Lesson:

When the component will be replaced soon, it may be more sensible to perform a minimal viable repair rather than a full rehabilitation.

Thematic Expansion — Kitchen Leaks and Progressive Renovations (November 2026)

Question: "I have broken pipes in my kitchen. What should I do?"

If there is an active leak, close the accessible supply when it can be done safely, protect the area, and contact a competent professional. If there is proximity to water and electricity, avoid dangerous manipulation. Once the fault is repaired, check for watertightness, possible moisture damage, and drying before restoring finishes. Do not claim that the entire installation is obsolete without inspection.

Question: "I also want to renovate the kitchen. What do you recommend?"

Take the opportunity to plan installations before closing walls, but separate essential repairs from finishes and optional furniture. Review the location of the sink, outlets, drains, and appliances; estimate by work item according to measurements and actual condition. It may be viable to retain the layout and sound components.

Question: «Can I also replace the tiles?»

Yes. As an optional part of the renovation, after resolving leaks and checking the substrate, tiles may be removed and replaced. Do not install coverings on damp or unstable substrates without proper treatment. Plumbing repair and wall finishes are separate tasks that must be budgeted separately.

Scope

These guidelines are general advice supported by guides/R4R_80_KITCHEN_PLUMBING_AND_LEAKS.md; they are not a new completed project or a real budget.